SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Chat Application

1. Introduction

Communication plays an important role in everyday life. With the rapid growth of internet technology, chat applications have become one of the most popular methods of communication. The purpose of this project is to develop a Chat Application that enables users to exchange messages instantly through an easy-to-use interface. The application is designed to provide real-time communication, user authentication, and secure message exchange.

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements of the Chat Application. It serves as a guideline for the development and maintenance of the system.

2. Purpose of the System

The main purpose of the Chat Application is to provide a platform where users can communicate with each other in real time. The system allows users to register, log in, send messages, receive messages, and manage their profiles. The application aims to provide a simple, reliable, and user-friendly communication experience.

3. Scope of the Project

The Chat Application is intended for students, teachers, employees, and general users who need an instant communication platform. The system supports user registration, login authentication, one-to-one messaging, profile management, and message history. Future enhancements may include group chats, voice calls, video calls, and file sharing features.

4. Overall Description

The Chat Application is a client-server-based system. Users interact with the application through a graphical user interface. The server processes requests, stores user information, and manages message delivery. The system ensures that messages are delivered efficiently and securely.

The application is designed to be accessible, easy to use, and responsive. Users can communicate anytime and from anywhere with an internet connection.

5. Functional Requirements

The Chat Application shall provide the following functionalities:

- User Registration
- User Login and Logout
- User Authentication
- Profile Management
- Sending Messages
- Receiving Messages
- Viewing Chat History
- Online/Offline Status Display
- Password Management

The system shall validate user information before account creation. Registered users shall be able to log in using valid credentials. Users can send and receive messages instantly. The system shall store chat history for future reference.

6. Non-Functional Requirements

The system shall satisfy the following non-functional requirements:

Performance

The application should respond quickly and deliver messages with minimal delay.

Reliability

The system should operate continuously without frequent failures.

Security

User credentials and messages should be protected from unauthorized access.

Usability

The interface should be simple, attractive, and easy to understand.

Scalability

The system should support an increasing number of users without affecting performance.

7. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 500 GB Hard Disk or SSD
- Internet Connection

Software Requirements

- Operating System: Windows/Linux/Android
- Programming Language: Java
- Database: MySQL
- IDE: VS Code / Eclipse / IntelliJ IDEA

8. Assumptions and Constraints

The application requires a stable internet connection for communication. Users must create an account before accessing chat features. The system performance depends on network availability and server capacity.

9. Conclusion

The Chat Application is designed to provide efficient and secure real-time communication among users. The system offers essential messaging features while maintaining simplicity and usability. This project demonstrates the practical implementation of software engineering concepts and serves as a foundation for future enhancements such as group chats, multimedia sharing, and voice communication.